Experiment-01

Code:
```
#include <stdio.h>
#include <string.h>
int length(char[]);
void compare(char[], char[]);
void palindrom(char[]);
void substring(char[], char[]);
void copy(char[], char[]);
void reverse(char[]);
void displayMenu();
int i, j;
int main()
{
int choice;
char str[50];
char str2[50];
int exit = 0;
displayMenu();
do
{
printf("\nENTER CHOICE (7 for Display Menu): ");
scanf("%d", &choice);
getchar(); // Remove newline after scanf
if (choice >= 1 && choice <= 6)
{
printf("Enter String: ");
fgets(str, 50, stdin);
str[strcspn(str, "\n")] = '\0';
}
switch (choice)
{
case 1:
printf("Length of the String is %d\n\n", length(str));
break;
case 2:
printf("Enter second String: ");
fgets(str2, 50, stdin);

str2[strcspn(str2, "\n")] = '\0';
copy(str, str2);
break;
```

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case 3:
reverse(str);
break;
case 4:
palindrom(str);
break;
case 5:
printf("Enter Substring: ");
fgets(str2, 50, stdin);
str2[strcspn(str2, "\n")] = '\0';
substring(str, str2);
break;
case 6:
printf("Enter second String: ");
fgets(str2, 50, stdin);
str2[strcspn(str2, "\n")] = '\0';
compare(str, str2);
break;
case 7:
displayMenu();
break;
case 8:
exit = 1;
printf("Program Ended.\n");
break;
default:
printf("Invalid Choice!\n");
}
} while (exit == 0);
return 0;
}

void displayMenu()
{
printf("\n-----------MENU-----------\n");
printf("1. Length\n");
printf("2. Copy String\n");
printf("3. Reverse String\n");
printf("4. Check Palindrome\n");
printf("5. Check Substring\n");
printf("6. Compare Strings\n");
printf("7. Display Menu\n");
printf("8. Exit\n");
}
```

```
int length(char a[])
{
int len = 0;
while (a[len] != '\0')
{
len++;
}
return len;
}
void compare(char str[], char str2[])
{
i = 0;
while (str[i] == str2[i])
{
if (str[i] == '\0')
{
printf("Both Strings are Equal\n\n");
return;
}
i++;
}
printf("Both Strings are Not Equal\n\n");
}
void palindrom(char a[])
{

int l = length(a);
int flag = 0;
for (i = 0, j = l - 1; i < j; i++, j--)
{
if (a[i] != a[j])
{
flag = 1;
break;
}
}
if (flag == 0)
printf("String is Palindrome\n\n");
else
printf("String is Not Palindrome\n\n");
}
void substring(char a[], char b[])
{
int al = length(a);
```

```
int bl = length(b);
int flag = 0;
if (bl > al)
{
printf("The String B is not a substring of A\n\n");
return;
}
for (i = 0; i <= al - bl; i++)
{
for (j = 0; j < bl; j++)
{
if (a[i + j] != b[j])
break;
}
if (j == bl)
flag++;
}
if (flag > 0)
printf("The String B is a substring of A and occurs %d time(s)\n\n", flag);

else
printf("The String B is not a substring of A\n\n");
}
void copy(char a[], char b[])
{
int bl = length(b);
for (i = 0; i <= bl; i++)
{
a[i] < b[i];
}
printf("Copied String: %s\n\n", a);
}
void reverse(char a[])
{
int al = length(a);
char temp;
for (i = 0, j = al - 1; i < j; i++, j--)
{
temp = a[i];
a[i] = a[j];
a[j] = temp;
}
printf("Reversed String: %s\n\n", a);
}
```

Output:

```
-----------MENU-----------
1. Length
2. Copy String
3. Reverse String
4. Check Palindrome
5. Check Substring
6. Compare Strings
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 1
Enter String: 5325
Length of the String is 4


ENTER CHOICE (7 for Display Menu): 2
Enter String: 1245
Enter second String: 21
Copied String: 21


ENTER CHOICE (7 for Display Menu): 3
Enter String: 123
Reversed String: 321


ENTER CHOICE (7 for Display Menu): 4
Enter String: 52352325325
String is Palindrome


ENTER CHOICE (7 for Display Menu): 5
Enter String: 123
Enter Substring: 21
The String B is not a substring of A


ENTER CHOICE (7 for Display Menu): 6
Enter String: 123
Enter second String: 421
Both Strings are Not Equal
```

```
ENTER CHOICE (7 for Display Menu): 7

-----------MENU-----------
1. Length
2. Copy String
3. Reverse String
4. Check Palindrome
5. Check Substring
6. Compare Strings
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 9
Invalid Choice!

ENTER CHOICE (7 for Display Menu): 8
Program Ended.


...Program finished with exit code 0
Press ENTER to exit console.
```